SUPPORTING ADULT GUIDANCE

Putting it together

Introduction to Trigonometry. For any supporting adult — teacher, practitioner, parent or carer. **You do not need to be a mathematics specialist to support this lesson.**

What this lesson does

The learner consolidates the whole unit and, above all, practises **diagnosis** — reading a right-angled-triangle problem and deciding which method it needs before calculating. Two questions frame every problem: **am I finding a side or an angle?** (angle given + side missing → a side; two sides given + angle missing → an angle), and **which ratio do the sides make?** (opposite & hypotenuse → sine; adjacent & hypotenuse → cosine; opposite & adjacent → tangent). The method then follows: a side is found by writing the ratio, substituting, rearranging (multiply if the unknown is in the numerator, divide if it is in the denominator) and evaluating; an angle is found by writing the ratio, substituting and applying the inverse. The lesson mixes both kinds so the learner cannot run one method on autopilot, and asks them to judge complete worked solutions for the unit's classic errors.

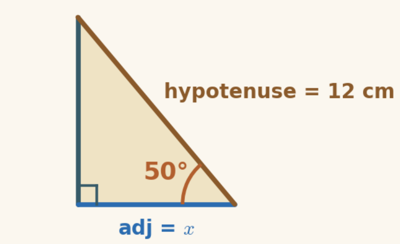
Driving Question: *Where does right-triangle trigonometry show up, and how do I choose my method?*

The mathematics, in plain terms

This is the unit's consolidation lesson, and its real subject is **choosing a method** — the half of mathematics that is easy to overlook when techniques are taught one at a time. A learner can often find a side and find an angle in isolation yet freeze when a problem does not announce which it is. The cure is a habit: ask two questions before calculating. **Question one — a side or an angle?** Look at what is unknown. If the angle is given and a length is missing, it is a side; if two lengths are given and the angle is missing, it is an angle. **Question two — which ratio?** Name the two sides in play and pick the ratio that uses exactly those two (SOH, CAH, TOA). Once both are answered the method is automatic: for a side, write the ratio, substitute, rearrange and evaluate — multiplying if the unknown sits on top, dividing if it sits underneath; for an angle, write the ratio, substitute and apply the inverse ($\sin^{-1}$, $\cos^{-1}$, $\tan^{-1}$) in degrees. The lesson deliberately shuffles side and angle problems together, and includes a 'spot the flaw' activity where the learner judges finished solutions — catching a wrong ratio, a multiply-when-you-should-divide, or a forgotten inverse. Judging others' work is one of the most efficient ways to sharpen the checking a learner then applies to their own. The real-world problems (a ladder against a wall, a ramp, a roof's pitch, a zip-wire's drop) show that the same two questions reach into the built world. Keep the calculator in degrees, and encourage the learner to say the diagnosis aloud before touching a key.

The worked method the learner sees

These are the two worked examples the learner meets on screen, with the lesson's exact method and notation. They are here for consistency, not because the mathematics is demanding — whether you teach mathematics or are meeting it afresh alongside the learner, working from the same steps and the same language keeps your support aligned with what is on the screen, and lets you point back to a line the learner has already seen. Skip them freely if the method is already familiar.

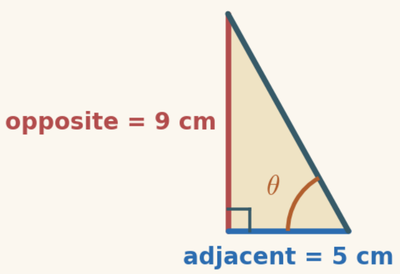


Example 1 — diagnosing a side problem

The angle (50°) is given and the adjacent side is missing — so this is a finding-a-side problem. The adjacent and hypotenuse go together, so use cosine.

1. a side or an angle?	a side
2. which ratio? (adjacent & hypotenuse)	$\cos 50° = \frac{x}{12}$
3. rearrange (x is on top, so multiply)	$x = 12 \times \cos 50°$
4. work it out	$x = 12 \times 0.643$

x = 7.71 cm



Example 2 — diagnosing an angle problem

This time two sides are given and the angle is missing — so this is a finding-an-angle problem. The opposite and adjacent go together, so use the tangent, then its inverse.

1. a side or an angle?	an angle
2. which ratio? (opposite & adjacent)	$\tan \theta = \frac{9}{5} = 1.8$
3. apply the inverse (undo tan)	$\theta = \tan^{-1}(1.8)$
4. work it out (degrees)	$\theta = 60.9°$

θ = 60.9°

Diagnose before you calculate

The organising idea of this lesson is **diagnosis** — and it is worth naming explicitly, because it transfers far beyond trigonometry. Skilled problem-solvers spend their first effort deciding what kind of problem they face; only then do they choose a tool. Two supports make the habit stick. First, insist on the **two questions in order** — side or angle? then which ratio? — every single time, even when the answer feels obvious; the routine is what carries a learner through the problems that are not obvious. Second, keep the **method visible**: once diagnosed, write every line, so a wrong turn shows itself. Mixing side and angle problems, as this lesson does, is essential — it removes the crutch of knowing in advance which method to run. Judging complete solutions for flaws builds the same muscle from the other side: a learner who can reliably spot a forgotten inverse or a multiply-that-should-be-a-divide in someone else's work is far more likely to catch it in their own. Finishing with real objects keeps the whole unit anchored to a simple truth — a right-angled triangle and two facts are enough to find the third.

The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

Cornerstone	Its role here	Invitational timing
Connection	Side or angle?: diagnose each problem — side-or-angle, then the ratio — and reveal the full method.	~8 min
Movement	Trig Problem Studio: diagnose and solve mixed side and angle problems unaided.	~12 min
Reflection	Spot the flaw: judge complete worked solutions for the unit's classic errors (dominant).	~14 min
Creativity	Reveal it in the world: solve real objects — ladder, ramp, roof, zip-wire — choosing the method.	~6 min
Rest	A pause after seeing the whole unit join into one habit.	~3 min
Nutrition	Mode B (intellectual): what the learning feeds — the skill of diagnosis, choosing a method before running it.	~2 min

Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

Running a method without diagnosing first

Repair: “Ask the two questions first — a side or an angle? which ratio? — before you reach for the calculator.”

Multiplying when the unknown is in the denominator

Repair: “Where is x? On top → multiply. Underneath (in the denominator) → divide. sin 30° = 5/x gives x = 5 ÷ 0.5 = 10.”

Forgetting the inverse when finding an angle

Repair: “A ratio like 0.75 is not the angle. Apply the inverse: $\theta = \tan^{-1}(0.75) = 36.9°$.”

Choosing the wrong ratio for the two sides

Repair: “Name the two sides in play, then pick the ratio that uses exactly those two — SOH, CAH, TOA.”

Calculator in radians, not degrees

Repair: “Check the mode says DEG — sin 30° should give 0.5 and $\sin^{-1}(0.5)$ should give 30.”

Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

Please hold this line

Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

Reconnection Routes

If an earlier idea is blocking progress, the lesson’s Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: SOH · CAH · TOA and choosing a ratio (Lesson 4), finding a side by writing and solving the ratio equation, including when the unknown is in the denominator (Lessons 5–6), and finding an angle with the inverse ratios (Lesson 7). A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

Supporting through questions: an example

Keep the learner diagnosing before calculating:

- “Is the angle given, or are you looking for it? So — a side or an angle?”
- “Which two sides are in play? So which ratio do they make?”
- “Is the unknown on top or in the denominator? So do you multiply or divide?”
- “A ladder against a wall — is that a side or an angle? How do you know?”

If they stall, that is the moment for a Reconnection Route or the lesson’s prompt and hint — not for the answer.

Questions you might have

“How do I know whether to find a side or an angle?”
Look at what is missing. If the angle is given and a length is unknown, you are finding a side. If two lengths are given and the angle is unknown, you are finding an angle. That single question chooses your path.

“Once I know it's a side, how do I know whether to multiply or divide?”
Write the ratio and substitute, then look at where the unknown sits. If it is in the numerator (on top), multiply. If it is in the denominator (underneath), divide.

“There are so many cases — is there one thing to remember?”
Yes: ask the two questions, in order, every time. Side or angle? Which ratio? Everything else — multiply or divide, inverse or not — follows from those two answers, so you never have to memorise a long list of separate rules.

Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

Recognising progress

Progress is described as Emerging → Developing → Secure → Mastering across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

Where they are	What it can look like
Emerging	Beginning to engage; names some features and makes a start with support.

Developing	Carries out the method with prompts; can explain part of the reasoning.
Secure	Works through independently and explains why each step is true.
Mastering	Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else.

A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.

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